Probing Fibronectin Type III Repeats Interaction with Synthetic Surfaces

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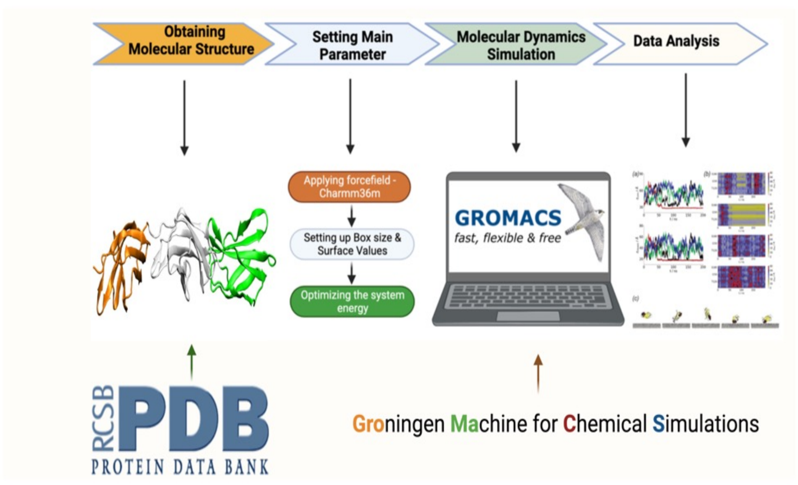
STATE-OF-THE-ART

Understanding protein-surface interactions at the molecular level is crucial for biomaterial science. Surface chemistry affects cellular responses like adhesion, survival, and inflammation. For example, anionic and neutral hydrophilic surfaces increase macrophage/monocyte death and reduce macrophage fusion, controlling inflammation from implants. These effects depend on how proteins interact with the surface, but the precise mechanisms are unclear. This limits the design of materials that elicit specific cellular responses. Molecular dynamics simulations can bridge this gap by providing detailed insights into protein adsorption and behavior on surfaces. This knowledge can lead to new biomaterials with tailored properties for improved biomedical device performance[1,2].

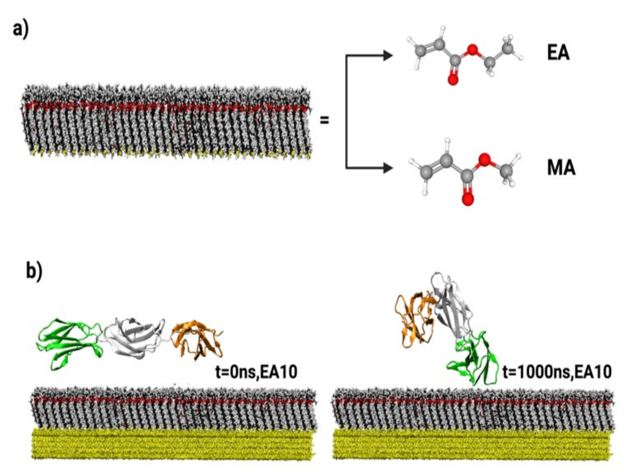
HYPOTHESIS

Minor changes in surface chemistry and protein sequence lead to significant alterations in protein adsorption. To investigate this, we hypothesize that molecular dynamics simulations will reveal the molecular factors controlling protein behavior at surfaces. By combining these simulations with experimental studies, we aim to understand protein behavior across different lengths and timescales. This approach will provide comprehensive insights into how surface properties influence protein adsorption and activity, ultimately guiding the design of biomaterials with tailored surface properties for improved biomedical applications.

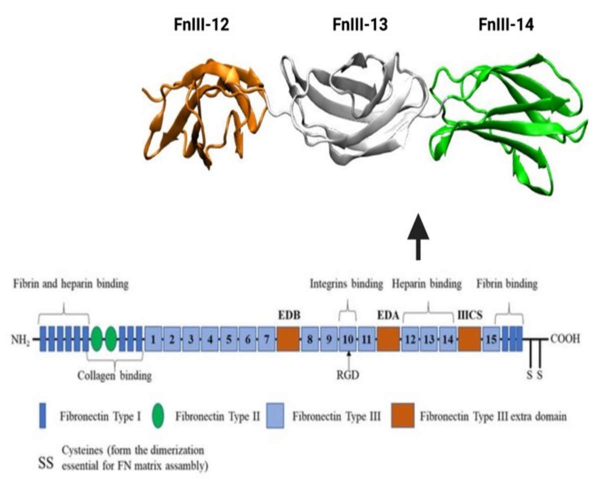
RESEARCH METHODOLOGY



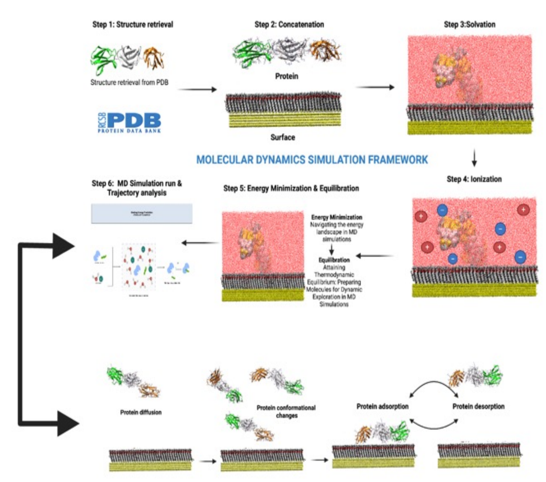
Research Outline



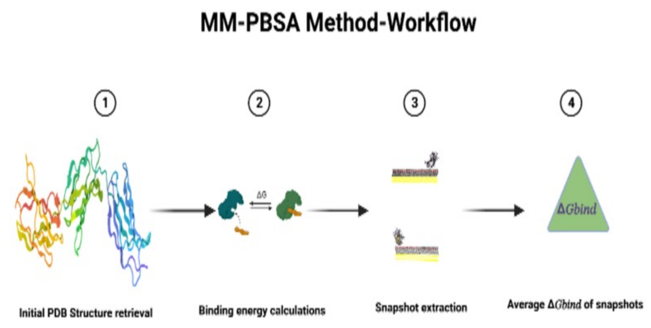
Surface Setup



Structure retrieval



Molecular dynamics Simulation framework



MM-PBSA Calculations

RESULTS & DISCUSSION

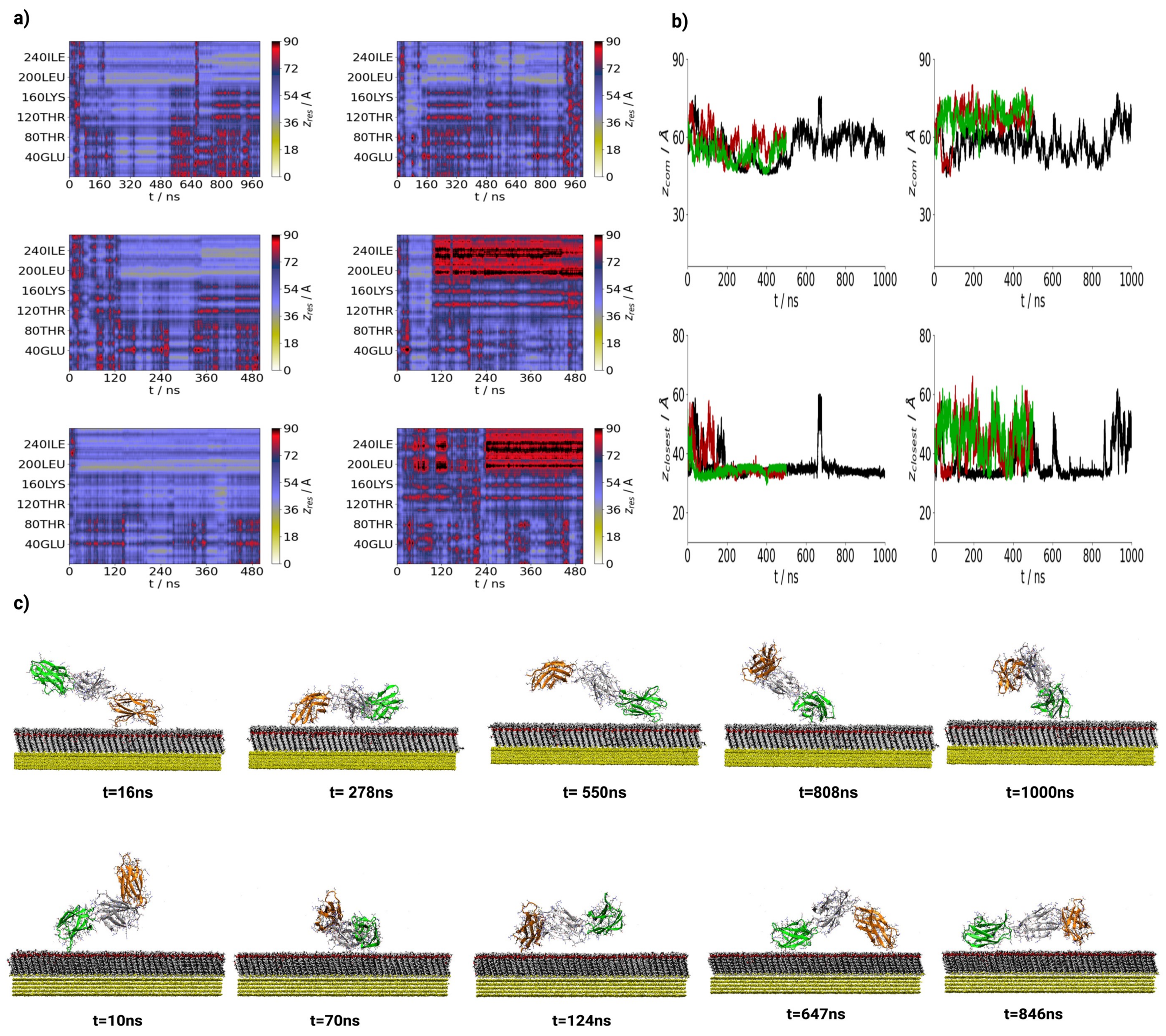


Figure 1: a) Residue *z* positions against time for Trimer on **EA10** [Left] & **MA10** [Right](b) Protein centre-of-mass (top: **EA10**-Left & **MA10**-Right) and closest residue (bottom:**EA10**-Left & **MA10**-Right) z-coordinates for trimer. **Black**, **red** and **green** denote first, second & third simulation runs, respectively. (c) Simulation snapshots taken from trimer on **EA10** [top: Run 1] & **MA10** [Bottom: Run 1]

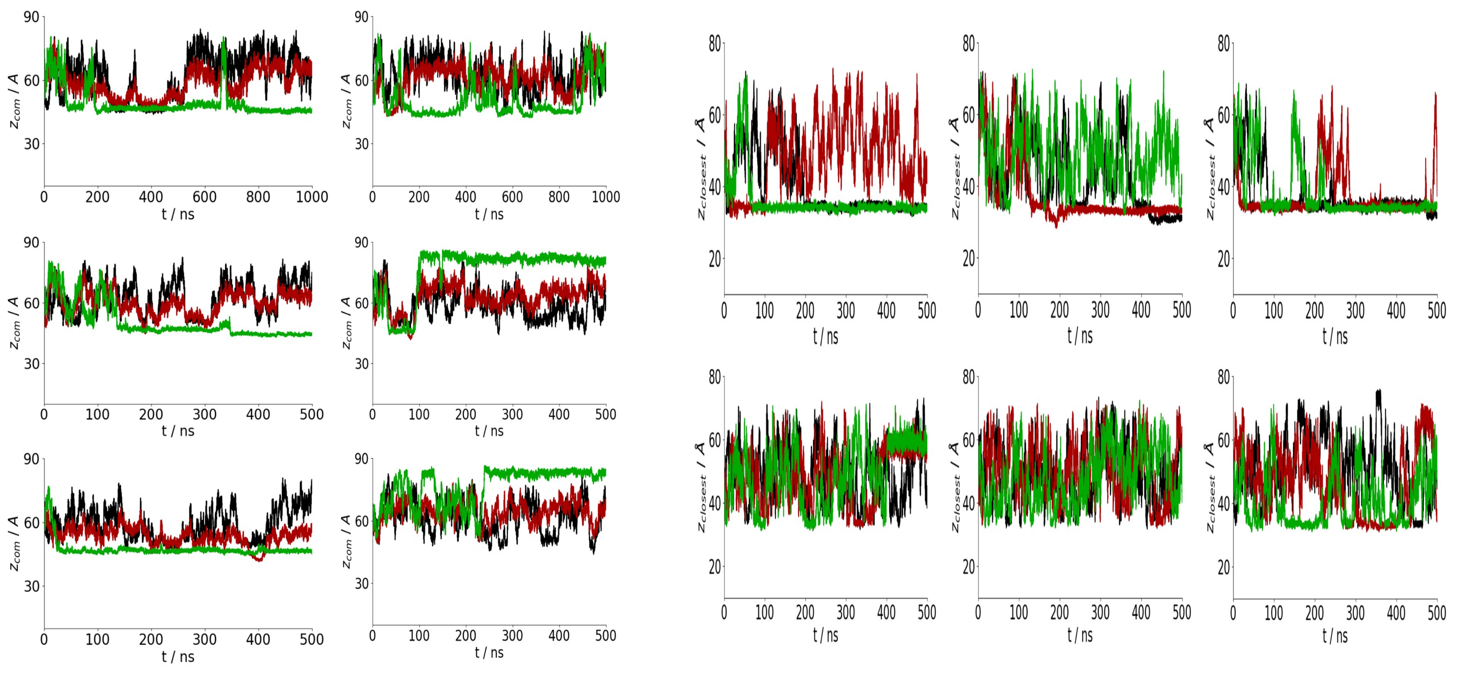


Figure 2: Protein centre-of-mass (**EA10**-Left & **MA10**-Right) and closest residue (top:**EA10**; bottom: **MA10**) z-coordinates for trimer. **Black**, **red** and **green** denote first, second & third simulation runs, respectively.

- For trimer simulations [Figure 1:a] adsorption was more stable on EA10 compared with MA10 surface. On EA10 surface adsorption was seen for longer periods for all runs whereas on MA10 only during run one and for other two runs, showed transient adsorption. Protein Centre of mass positions[Figure 1: b] for both surfaces and closest residues [Figure 1: c] indicated more favorable adsorption on EA10 compared with MA10 surface
- For single domain simulations [Figure 2] FnIII-12 ,FnIII-13 & FnIII-14 showed stable adsorption irrespective of their runs on SAM-EA10 surface, but it was only FnIII-14 on SAM-MA10 indicated adsorption. While during trimer simulations, both on SAM-EA10 & SAM-MA10 surfaces, the adsorption was largely due to FnIII-14 fragment indicating FnIII-12&FnIII-13 exhibits less affinity towards the surfaces. Protein centre of mass positions and closest residue plots clearly indicates the order of affinity towards surface is more for FnIII-14 followed by FnIII-13 & FnIII-12.

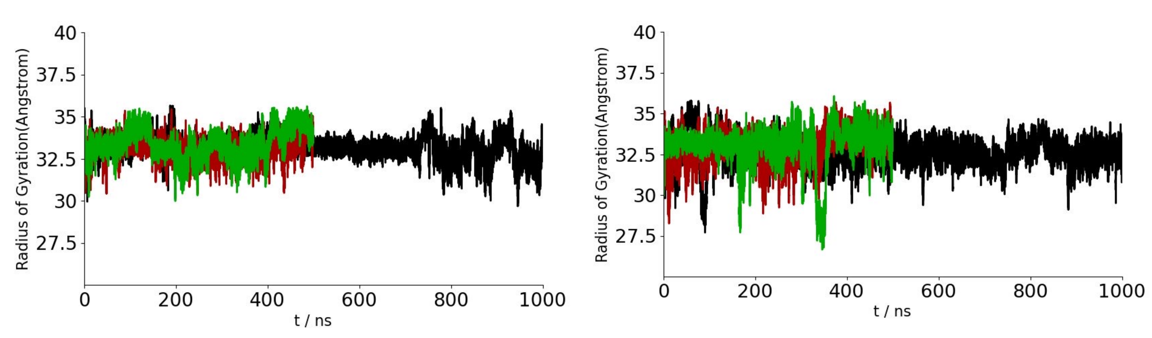


Figure 3: Radius of gyration for FnIII-9-10 trimer on EA10 (top) and MA10 (bottom) surfaces. Black, red, and green lines denote runs 1, 2, and 3 respectively.

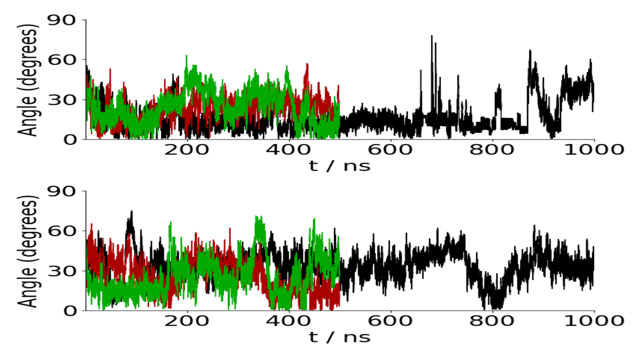


Figure 4:Interdomain angle (define this in text) for FnIII-9-10 trimer on EA10 (top) and MA10 (bottom) surfaces. Black, red, and green lines denote runs 1, 2, and 3 respectively.

CONCLUSIONS

Both from trimer and single domain simulations, similar residues showed contact onto EA10 surface irrespective of the runs and indicated stable adsorption for a period of over 20ns. From single domains, amino acid residues of fnIII-14 domains exhibited higher percentage of similarity with trimer in terms of contact with the surface followed by fnIII-12 & fnIII-13. In case of MA10 surface, mostly fnIII-14 single domain simulations indicated higher percentage of similarity with trimer in terms of amino acid residues contacted the surface followed by fnIII-12. Unlike EA10, fnIII-13 domains showed no contact with the surface both on single domains and trimer simulations.

COLLABORATIONS

Aston University BIRMINGHAM UK · University of Glasgow · UNIVERSITY OF BIRMINGHAM · Durham University · CCP5 · ICHEC Irish Centre for High-End Computing

EDUCATION & PUBLIC ENGAGEMENT

debra for people whose skin doesn't work

ICHEC Irish Centre for High-end Computing

OUTPUTS

ROYAL SOCIETY OF CHEMISTRY · biophysica · ESB2023 · ECMage · lifETIME Engineering Underpinning Medicine

REFERENCES

1. Bieniek, M. et al. Advanced Theory & Simulations. 2019, 2: 1900169.
2. Viswanath, V. et al. Biophysica. 2023, 3: 3030027.

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